EE 316

Section 3

Pre Lab 11

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MOSFETs

Introduction:

This lab investigates the operating characteristics of an n-channel MOSFET by examining the relationship between drain current, drain-to-source voltage, and gate-to-source voltage. A MOSFET operates in the linear ohmic region, where it behaves like a voltage-controlled resistor, and the saturation region, where the drain current becomes relatively constant. The gate-to-source threshold voltage is the minimum voltage required to establish a conducting channel between the drain and source. Through circuit simulation in LTspice, the output and transfer characteristics of a 2N7000 MOSFET are analyzed to identify these operating regions and determine its threshold voltage.

Simulations and Results:

The MOSFET circuit shown in Figure 1 was constructed and simulated in LTspice, as shown in Figure 2. Two simulations were performed to examine the MOSFET's output and transfer characteristics.

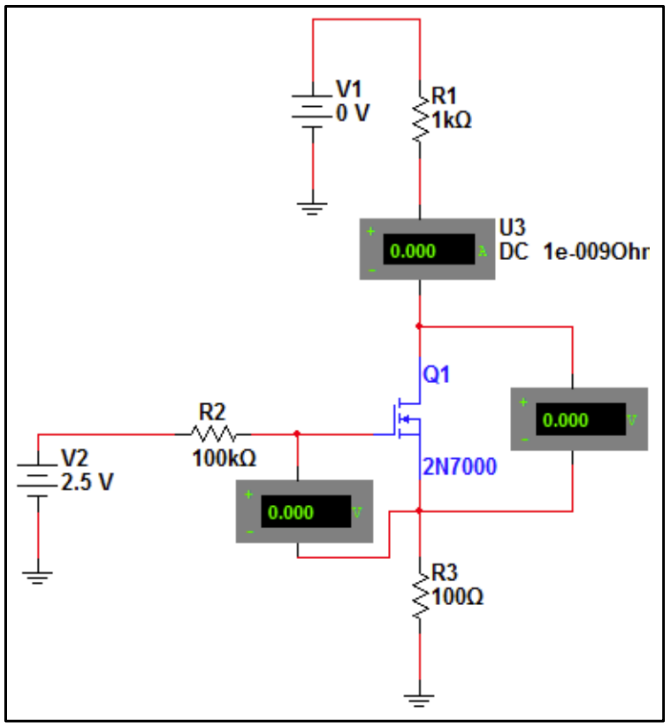


Figure #1: MOSFET Circuit Schematic (Lab Manual)

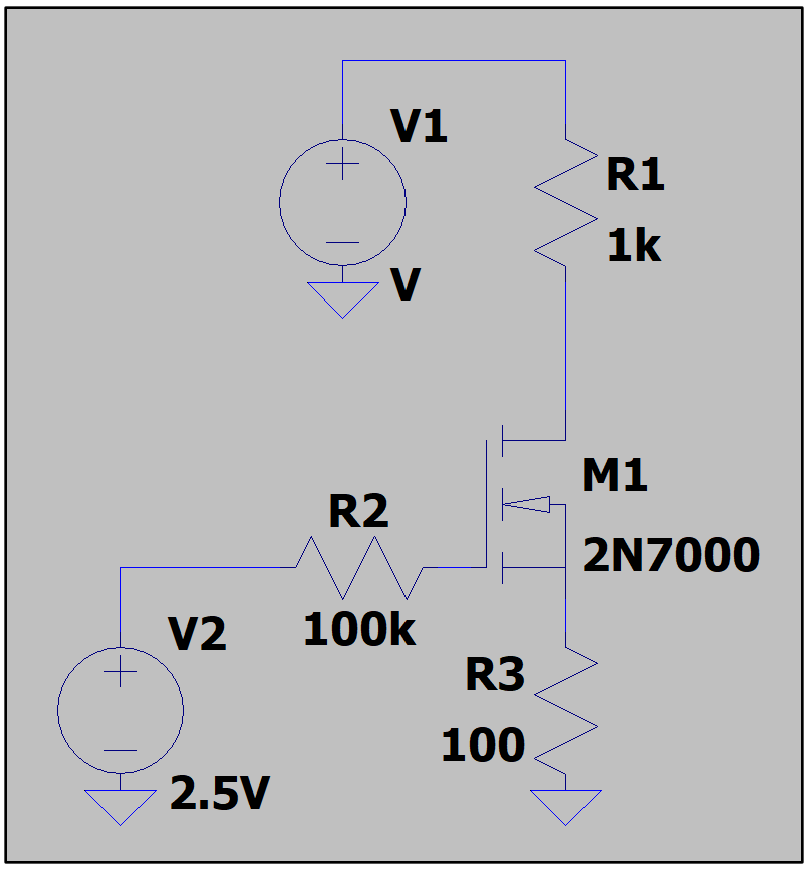


Figure #2: MOSFET Circuit (LTSpice)

For Part 1, V_2 was set to 2.5 V, and V_1 was increased from 0 V while measuring V_{GS}, V_{DS}, and I_D. The process was repeated with V_2 set to 3 V, 3.5 V, and 4 V. The results were recorded in Table 1 and plotted to show the relationship between drain current and drain-to-source voltage for each gate-to-source voltage. The plot in Figure 3 was used to identify the linear ohmic region, where the MOSFET behaves like a voltage-controlled resistor. The minimum V_{DS} required for saturation at each V_{GS} was also recorded based on the values in Figure 3.

$V_2 = 2.5$ V		$V_2 = 3$ V		$V_2 = 3.5$ V		$V_2 = 4$ V	
V_{DS}	$V_{GS} =$ 1.876 V	V_{DS}	$V_{GS} =$ 1.958 V	V_{DS}	$V_{GS} =$ 2.027 V	V_{DS}	$V_{GS} =$ 2.089 V
(mV)	I_D (mA)	(mV)	I_D (mA)	(mV)	I_D (mA)	(mV)	I_D (mA)
0	0	0	0	0	0	0	0
5.956	0.903	5.785	1.296	3.493	1.036	4.949	1.727
13.316	1.806	33.984	5.358	38.428	7.678	30.568	8.136
22.760	2.707	62.184	7.421	73.362	10.758	64.192	12.666
35.549	3.604	90.383	8.593	149.054	13.468	101.019	15.290
54.334	4.496	137.383	9.631	189.811	14.077	152.256	17.194
86.479	5.376	165.182	9.872	207.278	14.252	198.690	18.104
115.481	5.804	184.382	10.122	230.568	14.428	233.916	18.523
181.184	6.194	259.581	10.392	259.680	14.577	297.962	18.940
632.641	6.243	334.780	10.421	376.128	14.721	344.396	19.066
1132.641	6.243	438.178	10.421	498.399	14.727	382.824	19.111

Table #1: MOSFET Output Characteristics

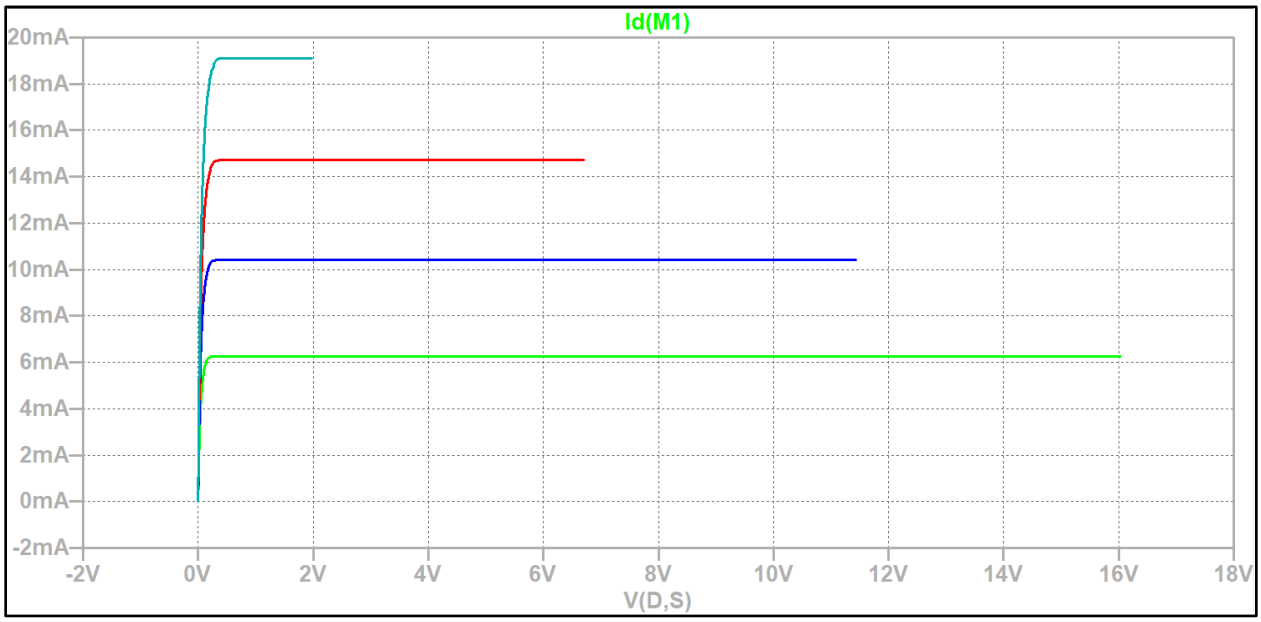


Figure #3: I_D vs. V_{DS} at

For Part 2, V_1 was set to 22 V, and V_2 was varied from 0 V in 0.25 V increments until the drain current stopped increasing. The corresponding V_{GS} and I_D values were recorded in Table 2 and plotted in Figure 4 to examine the MOSFET's transfer characteristics. The threshold voltage was identified from Figure 4 as the minimum gate-to-source voltage required for the MOSFET to begin conducting.

V_{GS} (V)	I_D (mA)
0	0.022
1	0.022
1.5	0.022
1.75	2.018
2	12.960
2.25	19.766
2.5	19.876
3	19.910

Table #2: MOSFET Output Characteristics

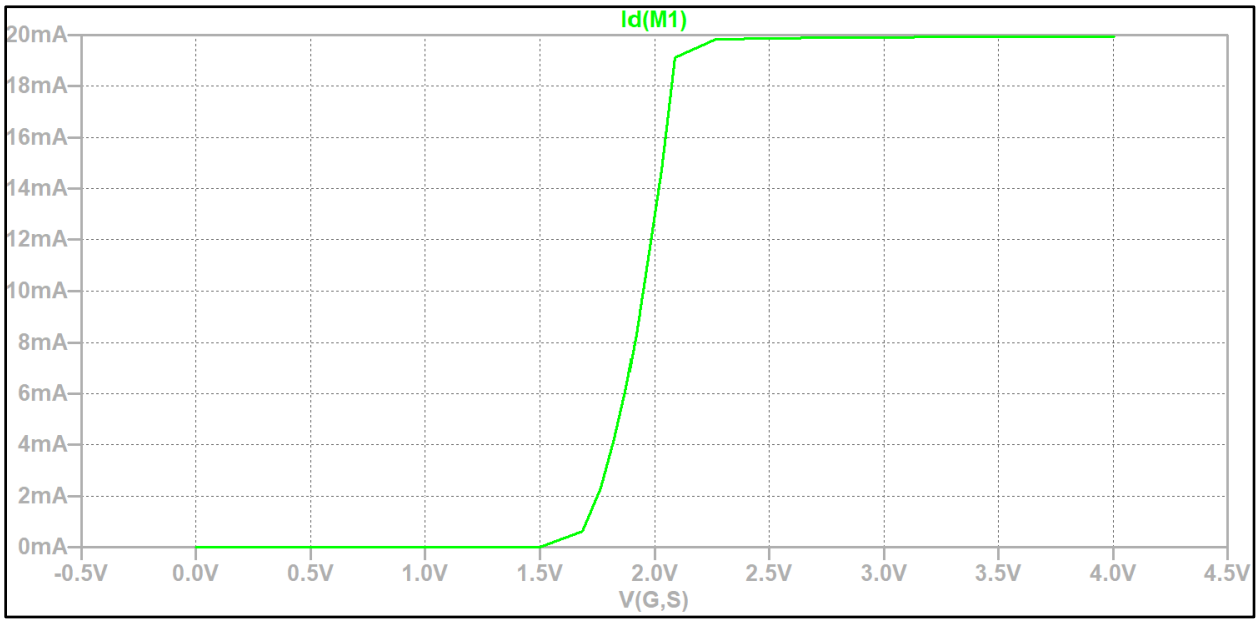


Figure #3: I_D vs. V_{DS}